NOT MEASUREMENT SENSITIVE

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MILITARY SPECIFICATION

ANTIFREEZE, ARCTIC-TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of antifreeze compound for use in the cooling system of liquid-cooled internal combustion engines for protection against freezing in ambient temperatures as low as -68 °C (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

O-C-265	- Chemicals, Analytical; General Specification for.
O-S-642	- Sodium Phosphate, Tribasic, Technical, Anhydrous, Dodecahydrate, and Monohydrate.
NN-P-71	- Pallet, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial).
UU-T-81	- Tag, Shipping and Stock.

Beneficial comments (recommendations, additions, deletions) and any pertintent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6850

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ZZ-H-428	- Hose, Rubber; Preformed and Straight; Hose, Preformed; Flexible, Wire-Reinforced (for the Cooling Systems of Automotive and Other Liquid-Cooled Engines).
PPP-B-601	- Box, Wood, Cleated Plywood.
PPP-B-636	- Box, Fiberboard.
PPP-C-569	- Containers, Plastic, Molded (for Liquids, Pastes and Powders); Overpacked.
PPP-C-1337	- Containers, Metal with Polyethylene Inserts.
PPP-D-1860	- Drums, Plastic, Molded, Polyethylene.
PPP-T-60	- Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-C-5501	- Cap and Plug, Protective, Dust and Moisture Seal.
MIL-C-5501/7	- Cap and Plug, Protective, Dust and Moisture Seal (Cap, Plug, General Purpose).
MIL-F-16377	- Fixture, Lighting, and Associated Parts, Shipboard Use, General Specification for.
MIL-B-26701	- Bottles, Screwcap and Carboys, Polyethylene Plastic.
MIL-D-43703	- Drum, Shipping and Storage, Molded Polyethylene.

STANDARDS

FEDERAL

FED-STD-313	- Material Safety Data Sheets, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)

49 CFR 171-179	- Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Code of Federal Regulations (CFR) are for sale on a subscription basis by the: SUPT OF DCMNTS, GVT PRINTG OFC, WASHINGTON DC 20402.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 92 - Flash and Fire Points by Cleveland Open Cup.
D 445 - Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity).
D 929 - Borax.
D 1119 - Ash Content of Engine Antifreezes, Antirusts, and Coolants.
D 1120 - Boiling Point of Engine Coolants.
D 1121 - Reserve Alkalinity of Engine Antifreezes.
D 1177 - Freezing Point of Aqueous Engine Antifreeze Solution.
D 1238 - Flow Rate of Thermoplastics by Extgrusion Plastometer.
D 1287 - pH of Engine Antifreezes, Antirusts and Coolants.
D 1881 - Foaming Tendencies of Engine Coolants in Glassware.
D 2240 - Rubber Property-Durometer Hardness.
D 3634 - Trace Chloride Ion in Engine Coolants.
E 29 - Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications.

(Application for copies should be addressed to: AMERCN SCTY & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.6) in accordance with 4.5.

3.2 Material. Materials shall be as specified herein.

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TABLE I. Chemical composition.

Component	Composition by weight percent of total antifreeze	Applicable test paragraph
Borax (ASTM D 929)	2.50 ±0.20	4.7.2.1
Trisodium phosphate calculated as dodecahydrate	0.37 ±0.04	4.7.2.2
Sodium salt of mercaptobenzothiazole solution, 50% aqueous solution by weight	0.46 ±0.05	4.7.2.3
Dye	1/	2/

1/ The antifreeze color (fluorescent yellow or orange yellow) shall be comparable to that produced by adding 0.36 gram of Acid Red 87 (color Index no. 45380) dye per gallon of antifreeze compound (see 6.3).
2/ Test not available.

3.2.1 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from waste materials and reprocessed to become a source of raw material, as distinguished from virgin raw material. The components used in the antifreeze compound may be newly fabricated from recovered materials, provided the antifreeze produced meets all the requirements of this specification.

3.3 Physical requirements.

3.3.1 Boiling point. When determined as specified in 4.7.2.4, the boiling point of the undiluted antifreeze shall be not lower than 110 °C.

3.3.2 Flash point. When determined as specified in 4.7.2.5, the flash point of the undiluted antifreeze shall be not lower than 68 °C.

3.3.3 Freeze point. When determined as specified in 4.7.2.6, the freeze point of aqueous solutions of the antifreeze shall be as specified in table II.

TABLE II. Freeze point of aqueous solutions.

Concentration (% by volume)	Freeze point (min. °C)
90	Not above -51 °C
60	Not above -21 °C

3.3.4 Compatibility.

3.3.4.1 With water. The undiluted antifreeze shall be completely miscible with water, in all proportions. When tested as specified in 4.7.2.7.1, aqueous solutions of the compound in the concentrations specified in table II shall show no precipitation or liquid separation. A slight turbidity is allowed.

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3.3.4.2 With reference solutions. When tested as specified in 4.7.2.7.2, the solutions obtained by mixing 1 part by volume of the antifreeze with 1 part of each of the reference solutions specified in table III shall show no precipitation or liquid separation.

TABLE III. Composition, by weight, of the reference solutions.

Component	Reference solution, % A	B	C
Ethylene glycol 1/	54.95	54.95	54.95
Distilled or deionized water	31.90	31.90	31.90
Ethylene glycol monomethyl ether 2/	9.70	---	---
Ethylene glycol monoethyl ether 3/	---	9.70	---
Diethylene glycol monomethyl ether 4/	---	---	9.70
Sodium tetraborate decahydrate (Borax) 5/	2.50	2.50	2.50
Trisodium phosphate dodecahydrate 6/	0.40	0.40	0.40
Sodium salt of mercaptobenzothiazole solution 7/	0.50	0.50	0.50
Antifoaming agent 8/	0.05	0.05	0.05

1/ The ethylene glycol used in the reference solutions shall be analyzed reagent grade conforming the latest commercial practice.
2/ The ethylene glycol monomethyl ether used in the reference solutions shall be analyzed reagent grade procured under O-C-265 with requirements conforming to nonmonograph section, USP, latest edition.
3/ The ethylene glycol monoethyl ether used in the reference solutions shall be analyzed reagent grade procured under O-C-265 with requirements conforming to nonmonograph section, USP, latest edition.
4/ The diethylene glycol monomethyl ether used in the reference solutions shall be reagent grade conforming to the latest commercial practice.
5/ Borax used in the reference solutions shall conform to ASTM D 929.
6/ The trisodium phosphate dodecahydrate used in the reference solution shall conform to O-S-642.
7/ The sodium salt of mercaptobenzothiazole solution shall be a 50 percent aqueous solution by weight.
8/ The antifoaming agent shall be of the polyoxyalkylene glycol type such as Pluronic L-61 supplied by BASF, Wyandotte Inc., Wyandotte, MI, or equal.

3.3.5 Kinematic viscosity. When determined as specified in 4.7.2.8, the kinematic viscosity of the undiluted antifreeze shall be not more than 1750 centistokes at -54 °C.

3.3.6 Foaming. When tested as specified in 4.7.2.9, the undiluted antifreeze shall show a maximum of 150 mL of foam after 5 minutes and the foam shall break within 5 seconds.

3.4 Chemical requirements.

3.4.1 pH. When determined as specified in 4.7.2.10, the pH of the undiluted antifreeze, and of a 60 percent aqueous solution by volume of the antifreeze shall be within the range of 7.0 to 8.5.

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3.4.2 Reserve alkalinity. When determined as specified in 4.7.2.11, the reserve alkalinity of the undiluted antifreeze shall be within the range of 13.5 to 18.5.

3.4.3 Ash. When tested as specified in 4.7.2.12, the ash content of the undiluted antifreeze shall be within the range of 0.7 to 1.3 percent, by weight.

3.4.4 Alkaline earths. When tested as specified in 4.7.2.13, the undiluted antifreeze shall form no immediate precipitate.

3.4.5 Sulfates and carbonates. When tested as specified in 4.7.2.14, the undiluted antifreeze shall show no undissolved precipitate or effervescence. A slight turbidity is allowed.

3.4.6 Chloride. When tested as specified in 4.7.2.15, the undiluted antifreeze shall have a chloride content of 25 ppm or less.

3.4.7 Rubber softening. When tested as specified in 4.7.2.16, the rubber softening caused by the undiluted antifreeze shall not exceed the softening caused by at least 1 of the reference solutions (see table III) similarly tested.

3.4.8 Storage stability. After six months storage in accordance with 4.7.2.17, the undiluted antifreeze shall meet all workmanship requirements (see 3.9). In addition, the unit containers shall not show disintegration due to action of the antifreeze.

3.5 Toxicological product formulation. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertaining to the toxic effects shall be referred by the procuring activity to the appropriate departmental medical service who will act as an advisor to the procuring activity.

3.5.1 Hazardous material. The contractor shall comply with the hazardous material requirements of FED-STD-313 (see 6.5).

3.6 Transportability. Wherever and whenever applicable packaging, packing, and marking of the antifreeze shall be prepared for delivery in accordance with the requirements of the Code of Federal Regulations 49 CFR, Parts 171-179.

3.7 Transportation. The following applies to all specified limits in this specification: For purposes of determining conformance with these specifications, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right-hand digit used in expressing the specification limit in accordance with the rounding-off method of ASTM E 29.

3.8 Workmanship. The undiluted antifreeze shall meet all the chemical and physical requirements of this specification. In addition to these requirements, the undiluted antifreeze shall be translucent in appearance, but free of any insoluble suspensions, such as dirt, undissolved additive, or foreign matter. The undiluted antifreeze shall be free of visibly excessive amounts of antifoaming agent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor's own facilities or any other, suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Inspection lot. All compounded antifreeze of the same type manufactured as one batch and offered for delivery at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5).
b. Quality conformance inspection (see 4.6).
c. Inspection of packaging (see 4.8).

4.4 Toxicological product formulation review. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use through the submission of the Material Safety Data Sheet detailed in FED-STD-313 (see 3.6.1 and 6.5).

4.5 First article inspection.

4.5.1 First article examination. The first article shall be examined as specified in 4.7.1. One or more defects shall be cause for rejection.

4.5.2 First article tests. The first article shall be tested as specified in 4.7.2.1 through 4.7.2.17. Failure of any test shall be cause for rejection.

4.6 Quality conformance inspection.

4.6.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

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4.6.2 Examination. Samples selected as specified in 4.6.1 shall be examined as specified in 4.7.1. Presence of one or more defects shall be cause for rejection.

4.6.3 Tests. Samples selected as specified in 4.6.1 shall be tested as specified in 4.7.2.5, 4.7.2.6, 4.7.2.7, 4.7.2.8, 4.7.2.10, 4.7.2.11, 4.7.2.14 and 4.7.2.15. Failure of any test shall be cause for rejection.

4.7 Inspection procedure.

4.7.1 Examination. The antifreeze shall be examined as specified herein for the following defects:

101. Material not as specified (see 3.2).
102. Workmanship not as specified (see 3.8).

4.7.2 Tests.

4.7.2.1 Borax.

4.7.2.1.1 Procedure. Weigh approximately 100 grams of the antifreeze to the nearest 0.1 milligram into a 500-mL Erlenmyer flask. Dissolve in 100 mL freshly boiled distilled water. Add a five percent solution of $BaCl_2$ in slight excess while stirring. After settling for 30 minutes, filter the solution through a fine textured filter paper (Whatman 42 or equivalent) into a 500-mL volumetric flask. Wash the residue with two 10-mL portions of boiled distilled water. Return the paper to the original flask. Dissolve the residue in 6 N HCl and add 50-mL distilled water. Stir the sample to break up the paper. Reprecipitate the phosphate with 6 N NaOH. Add a few drops of sodium hydroxide and a few drops of barium chloride to check for completeness of precipitation. Filter and wash as before into the 500-mL volumetric flask. Cool the filtrate to 20 °C and dilute to the mark with cooled boiled distilled water. Take a 100-mL aliquot for analysis. Each aliquot contains one-fifth of the sample weight. Adjust the solution to pH 5.5 with 1N HCl using glass electrodes and instrument described in ASTM D 1121. Add 1 gram of mannitol and titrate to pH 8.5 with standardized 0.5 N NaOH. Add 0.5 grams of mannitol. If no change in pH is observed, the titration is complete. If a change is observed, add 0.5-gram portions of mannitol and subsequently titrate to pH 8.5 until no change in pH occurs. Calculate the percent $Na_2B_4O7{\cdot}10H_2O$. One milliequivalent of NaOH = 0.09537 grams of $Na_2B_4O_7{\cdot}10H_2O$. Nonconformance to table I shall constitute failure of this test.

4.7.2.1.1 Calculation of borax.

$$\text{Percent } Na_2B_4O_7 \cdot 10H_2O \text{ (borax)} = \frac{(mL\ NaOH)\ (47.685)}{\text{weight of original sample}}$$

4.7.2.2 Trisodium phosphate dodecahydrate.

4.7.2.2.1 Procedure. Accurately weigh approximately 50 grams of antifreeze to nearest milligram into a 600-mL beaker. Dissolve in 250 mL of freshly distilled water. Add 20 mL concentrated nitric acid and 20 grams ammonium nitrate to the

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beaker. Heat the solution to 30 - 45 °C. Add 50 mL ammoniacal molybdate solution (see 4.7.2.2.2). Do not heat the solution after the addition of the molybdate solution. Agitate the solution for 10 minutes, then allow to settle for about 30 minutes. Filter the solution through fine textured filter paper (Whatman 42 or equivalent) and rinse the beaker with five 20-mL portions of 1.0 percent potassium nitrate solution. Wash the paper with 100 mL of 1.0 percent potassium nitrate using small portions. Discard all filtrates. Return the paper and precipitate to the beaker and add standardized 0.5 N sodium hydroxide from a buret to decompose the precipitate. Add an excess of 2.0 mL. Record the volume of 0.5 N sodium hydroxide used. Wash the sides of the beaker and dilute to about 150 mL with distilled water. Titrate the resulting solution with standardized 0.1 N nitric acid to pH 8.5 using glass electrodes and instrument described in ASTM D 1121. Nonconformance to table I shall constitute failure of this test.

4.7.2.2.2 Ammoniacal molybdate solution. Weigh 61 grams ammonium molybdate and 57 grams ammonium nitrate into a one liter beaker. Add 47 mL concentrated ammonium hydroxide and 500 mL distilled water. Warm the beaker to aid dissolution. Filter and make up to one liter in a volumetric flask.

4.7.2.2.3 Calculation of trisodium phosphate dodecahydrate.

$$\text{Percent } Na_3PO_4 \cdot 12H_2O = \frac{[(1.653)\ (mL\ NaOH\ X\ normality\ NaOH) - mL\ HNO_3\ X\ normality\ NHO_3)]}{weight\ sample}$$

4.7.2.3 Sodium salt of mercaptobenzothiazole solution (NaMBT plus H_2O, 50 percent by weight).

4.7.2.3.1 Procedure. Accurately weigh 30 grams of antifreeze to the nearest milligram into a 250-mL beaker and dilute to 100 mL with freshly distilled water. Add 15 mL of a 10 percent copper sulfate solution and allow the mixture to stand for one hour. Add 5 mL of 1.0 N hydrochloric acid and stir. Digest the mixture at 50 °C for 30 minutes and let cool to room temperature. Filter the precipitate of copper mercaptobenzothiazole through a tared Gooch or sintered glass crucible, wash with 0.5 N hydrochloric acid until the precipitate is free of the blue color, then with distilled water until the washings are free from chloride. Dry the crucible in an oven at 105 to 120 °C to constant weight.

4.7.2.3.2 Calculation of sodium salt of mercaptobenzothiazole solution.

$$\%\ \text{sodium mercaptobenzothiazole solution} = \frac{192\ X\ \text{weight of precipitate}}{\text{weight of antifreeze sample}}$$

4.7.2.4 Boiling point. The boiling point of the undiluted antifreeze shall be determined in accordance with ASTM D 1120. Nonconformance to 3.3.1 shall constitute failure of this test.

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4.7.2.5 Flash point. The flash point of the undiluted antifreeze shall be determined in accordance with ASTM D 92. Nonconformance to 3.3.2 shall constitute failure of this test.

4.7.2.6 Freeze point. The freeze point of the aqueous solutions of the antifreeze (see table II) shall be determined in accordance with ASTM D 1177. Nonconformance with 3.3.3 shall constitute failure of this test.

4.7.2.7 Compatibility.

4.7.2.7.1 With distilled water. The aqueous solutions of the antifreeze used in the determination of the freeze point (see 4.7.2.6), shall be allowed to warm to room temperature (approximately 25 °C), without agitating the samples, and examined for appearance. The samples shall then be transferred to 125-mL oil sample bottles, stoppered with corks, and kept for 24 ±2 hours in an oven maintained at 82 ±3 °C. Exposure to light shall be kept to a minimum. At the end of this period, the samples shall again be examined for appearance. Nonconformance to 3.3.4.1 shall constitute failure of this test.

4.7.2.7.2 With reference solutions. One hundred milliliters of 1-to-1 mixtures (by volume) of the antifreeze and each of the reference solutions specified in table III (see 3.3.4.2) shall be placed in 125-mL oil sample bottles, and exposed for 24 ±2 hours at -68 °C. At the end of this period, the samples shall be examined for appearance. The samples shall then be allowed to warm to room temperature (approximately 25 °C) and again examined for appearance. The samples shall then be heated for 24 ±2 hours in an oven maintained at 82 ±3 °C. At the end of which they shall be again examined for appearance. During this procedure, exposure to light shall be kept to a minimum. Nonconformance to 3.3.4.2 shall constitute failure of this test.

4.7.2.8 Kinematic viscosity. The kinematic viscosity of the undiluted antifreeze shall be determined in accordance with ASTM D 445. Nonconformance to 3.3.5 shall constitute failure of this test.

4.7.2.9 Foaming test. The antifreeze shall be tested for foaming tendencies in accordance with ASTM D 1881. Nonconformance to 3.3.6 shall constitute failure of this test.

4.7.2.10 pH. The pH of the antifreeze shall determined in accordance with ASTM D 1287. Nonconformance to 3.4.1 shall constitute failure of this test.

4.7.2.11 Reserve alkalinity. The reserve alkalinity of the antifreeze shall be determined in accordance with ASTM D 1121. Nonconformance with 3.4.2 shall constitute failure of this test.

4.7.2.12 Ash. The ash content of the antifreeze shall be determined in accordance with ASTM D 1119. Nonconformance to 3.4.3 shall constitute failure of this test.

4.7.2.13 Alkaline earths. Place 10 mL of the material in a 30-mL test tube. Make alkaline to litmus paper with ammonium hydroxide solution. Add 5 mL of 95 percent ethyl alcohol reagent grade. Add 5 mL of 0.5 N ammonium oxalate. No immediate precipitate shall form. Nonconformance to 3.4.4 shall constitute failure of this test.

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4.7.2.14 Sulfate and carbonates. Place 10 mL of antifreeze in a 30-mL test tube. Add 10 mL of distilled water and mix. Add 2 mL of a 5 percent aqueous solution of barium chloride. A precipitate will form. Add 1 mL glacial acetic acid. There shall be no undissolved precipitate or effervescence. A slight turbidity is allowed. Nonconformance to 3.4.5 shall constitute failure of this test.

4.7.2.15 Chloride. The chloride content of the antifreeze shall be determined in accordance with ASTM D 3634. Nonconformance to 3.4.6 shall constitute failure of this test.

4.7.2.16 Rubber softening. The following materials, apparatus, and procedure shall be used in determining the rubber-softening property of the antifreeze compound and of the reference solutions.

4.7.2.16.1 Rubber hose. Three specimens of hose shall be used for each immersion test in the antifreeze compound and in the reference solutions. The hose shall conform to ZZ-H-428, type I, II or III, grade A, class 2 or 4. The outside diameter of the hose shall be approximately 51 mm. Each hose specimen shall consist of a 13-mm wide ring of hose, split crosswise so that it may be opened.

4.7.2.16.2 Procedure.

4.7.2.16.2.1 Hardness (Shore durometer). The hardness of each hose specimen shall be determined, before and after the immersion test, on the inside surface of the hose tube using the Shore A durometer apparatus detailed in ASTM D 2240.

4.7.2.16.2.2 Immersion. Place 3 specimens (split rings) of the rubber hose in each of four 1-pint glass jars having tight fitting lids (Mason jar type). Add 100 mL of the test antifreeze and each of the reference solutions (see table III) to separate jars. Tighten the lids on the jars and place in an oven maintained at 71 ±3 °C for 120 ±2 hours. At the end of the heating period, the hose specimens shall be removed from the jars, rinsed with tap water, followed by ethyl alcohol, and dried by blowing compressed air over them. A durometer reading shall be made on each specimen as soon as the specimens are dried. The durometer readings on the hose specimens from each reference solution shall be compared with those obtained on the specimens from the test antifreeze. Nonconformance to 3.4.7 shall constitute failure of this test.

4.7.2.17 Storage stability. Place one unit container furnished with contract or purchase order (see 6.6) in darkness, at 24 ±3 °C, under static storage conditions for six months. After six months, inspect unit container and antifreeze. Nonconformance to 3.4.8 shall constitute failure of this test.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of packing.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack, prepared for shipment shall be considered a unit of product.

4.8.1.2 Examination. Each unit product shall be examined for the following defects. One or more defects shall be cause for rejection:

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103. Preservation not as specified for level A, or commercial (see 5.1).
104. Container types or classes and sizes not as specified for level A (see 5.1).
105. Packing not as specified for level A, B, or commercial (see 5.2).
106. Shipping containers not as specified for level A, B, or commercial (see 5.2.1, 5.2.2, and 5.2.3).
107. Marking incorrect, illegible, or missing (see 5.3).
108. Radiator tags not as specified (see 5.4).
109. Palletization not as specified, when required (see 5.5).

5. PACKAGING

5.1 Preservation (unit packing). Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 One-gallon quantity. One US gallon of antifreeze shall be placed in a composite polyethylene-fiberboard container conforming to PPP-C-569, type I, class B, or in a one-gallon capacity polyethylene plastic bottle in accordance with MIL-B-26701. If the bottle specified in MIL-B-26701 is being used, an additive shall be used in the polyethylene resin (see 6.4) so that no portion of the bottle shall transmit more than 1 percent of ultraviolet light at any wavelength in the range of 310 to 325-millimicron when tested using a calibrated spectrophotometer with air as a reference. A leak-proof inner seal which shall prevent leakage of container contents without benefit of the screw-cap closure when the container is inverted; it shall be furnished beneath the screw-cap of each container.

5.1.1.2 Five-gallon quantity. A five US gallon of antifreeze shall be placed in a container or drum in accordance with PPP-C-1337, type II, class 1, or MIL-D-43703, size 1. A leak-proof, either inner seal or pilfer-proof locking seal shall be furnished with the screw-cap closure of each container or drum. The filled container or drum shall be closed tightly to prevent leakage of the contents.

5.1.1.3 Fifty-five gallon quantity. A fifty-five US gallon of antifreeze shall be placed in a container or drum in accordance with PPP-C-1337, type II, class 4, or PPP-D-1860. If the latter drum (as in PPP-D-1860) is used, the polyethylene shall have a specific gravity of 0.941 to 0.965 and a melting index of 0.20 to 1.2 gm per 10 minutes using procedure A, condition E, of ASTM D 1238. The filled container or drum shall be closed tightly with rigid plastic fittings to prevent leakage of contents.

5.2 Packing. Packing shall be level A, B, or commercial (see 6.2).

5.2.1 Level A.

5.2.1.1 One-gallon quantity. Six one-gallon containers preserved as specified in 5.1.1.1 shall be packed in a close fitting box in accordance with PPP-B-601, overseas type, class and style optional. Interlocking partitions, fabricated from fiberboard, shall be utilized to separate the containers as required.

5.2.1.2 Five-gallon and fifty-five-gallon quantities. The five- and fifty-five-gallon quantities shall require no further packing.

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5.2.2 Level B.

5.2.2.1 One gallon quantity. Packing shall be as specified in 5.2.1.1, except the exterior box shall be in accordance with PPP-B-636, class weather resistance, variety single wall, grade optional.

5.2.2.2 Five-gallon and fifty five-gallon quantities. The five and fifty five-gallon quantities shall require no further packing.

5.2.3 Commercial.

5.2.3.1 One-quart and one-gallon quantities. In the quantities shown (see 6.2), packing shall be as specified in ASTM D 3951.

5.2.3.2 Five-gallon and fifty five-gallon quantities. The five- and fifty-five gallon quantities shall require no further packing.

5.3 Marking. Marking shall be in accordance with MIL-STD-129. In addition, each container shall be marked to show the lot number and date of manufacture and the following:

"INSTRUCTIONS FOR USE"
For use at temperatures of -40 to -68 °C.
DO NOT DILUTE - USE AS PACKAGED.

Before installation, inspect cooling system, repair leaks and replace worn parts. If solution or system is dirty, flush thoroughly. Install antifreeze coolant as required (see instructions) to within 25 mm of overflow. Immediately (and before driving) run engine to normal driving temperature (or until thermostat opens) with heater on full. This releases trapped air. If necessary, add more antifreeze. Check solution level periodically. If level falls without visible leakage, check for internal leaks which may cause serious damage with any type of antifreeze coolant. Add antifreeze coolant as required.

WARNING

HARMFUL OR FATAL IF SWALLOWED. Do not drink antifreeze. If swallowed, induce vomiting immediately. Call a physician. Ethylene glycol base. Do not store in open or unlabeled containers. KEEP OUT OF REACH OF CHILDREN.

5.4 Radiator tags. Unless otherwise specified (see 6.2), radiator tags for the antifreeze shall be provided on, or with, containers of antifreeze in the following quantities: Not less than 4 tags or labels with each six 1-gallon containers or 5-gallon container and 40 tags or labels with each 55-gallon container. Pressure-sensitive labels may be used in lieu of radiator tags. Radiator tags shall conform to UU-T-81, type B, grade 20R, wire strung, size 5 and shall be marked on side 1 and 2 as shown on figure 1. Radiator labels shall conform to MIL-F-16377, except that the adhesive be water-insoluble, pressure-sensitive, and protected by release paper backing. These labels shall be 60 mm wide and 15 mm long ±2 mm and shall be marked as shown on figure 2. Tags or labels shall be packed together in a bag as specified in MIL-P-116, method IC-3. The bag shall be placed in the box or on the drum, as applicable. When placed in a box, the bag shall be placed flat on the top of the contents and beneath the box flaps. When placed on the drum, it shall be secured to the drum with tape

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conforming to PPP-T-60, type IV, class 1 in an area not normally exposed to abrasion.

5.5 Palletization. When specified (see 6.2), like packs of 1-gallon and 5-gallon quantities shall be palletized. Level A, B, and commercial packs shall be palletized in accordance with the applicable requirements, including those for opaque shrink film bonding of MIL-STD-147, using new, unused seasoned softwood or hardwood pallets conforming to NN-P-71, type IV. Polyethylene containers, with or without fiberboard overpack, shall be palletized using load type VI as specified in MIL-STD-147. The maximum number of 5-gallon drums placed on each pallet shall not exceed the weight and size limitations of the pallet. Excess shipping containers which do not constitute a full pallet course, but having a combined gross weight of 91 kilograms or more, shall be prepared for shipment in a manner assuring transport of the containers in a unitized state from initial source to ultimate activation. The containers shall be consolidated in such a manner that the load con be moved by forklift truck or other mechanical means. Excess shipping containers which do not constitute a full pallet course but having a combined gross weight of 90.5 kilograms or less may be consolidated as specified above or may be shipped as individual containers.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Arctic-type nonvolatile antifreeze is intended for use in the cooling system of liquid cooled internal combustion engines for protection against freezing in regions where the ambient temperature remains for extended periods of time close to -40 °C, but may drop as low as -68 °C.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article is not required (see 3.1).
d. Degree of preservation and packing required (see 5.1 and 5.2).
e. Unit quantities required (see 5.1.2 and 5.2.3.1).
f. Marking required (see 5.3).
g. When radiator tags are not required (see 5.4).
h. When palletization is required (see 5.5).
i. When a PIN number is required (see 6.9).

6.3 Color Index. The Color Index, published by the American Association of Textile Chemists and Colorists, lists the commercial names and manufacturers of dyes conforming to Acid Red 87 (Color Index No. 45380).

6.4 Additive to plastic containers. An additive which is known to be effective in excluding ultraviolet light from the interior of polyethylene containers is phthalocyanine blue.

6.5 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in

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accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313.

6.6 First article. When a first article inspection is required, the item should be an initial production sample. The first article should consist of four one-quart containers, when quart containers are specified, or not less than one gallon when all other containers are specified. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results, and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.7 Disposability. Large quantities of antifreeze concentrate may be disposed of by mixing the material with flammable solvents and atomizing the mixture into an incinerator. Contact a licensed contractor for detailed recommendations. Follow Federal, state, and local regulations.

6.8 Subject term (key word) listing.

Alaska
Coolant
Cooling system
Corrosion inhibitor
Engine, internal combustion
Heavy-duty diesel
Radiator

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 68

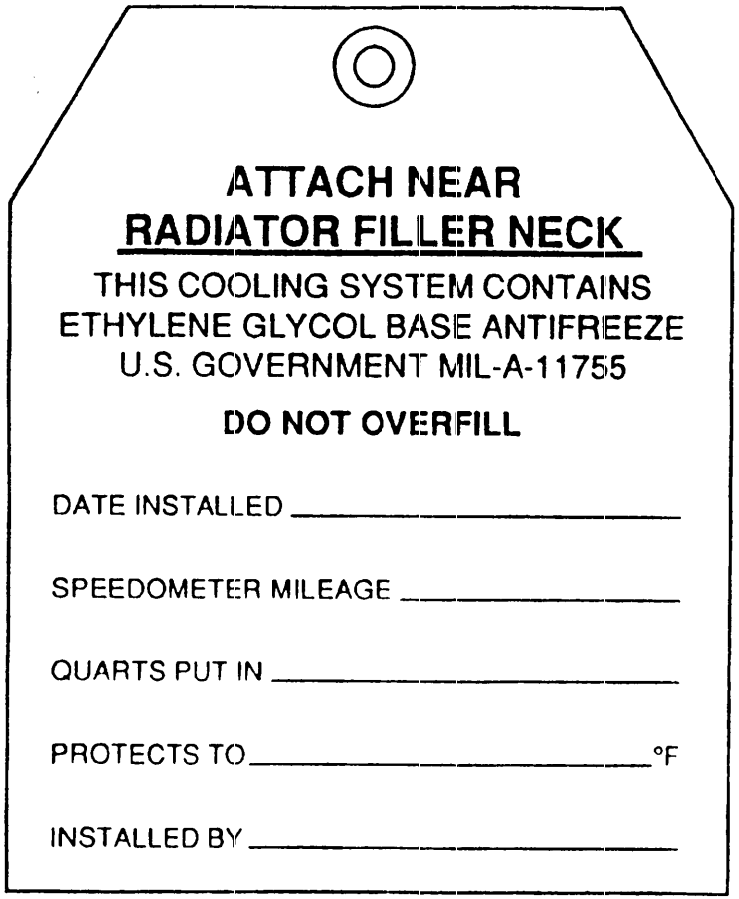
Preparing activity:
Army - ME

Project 6850-1046

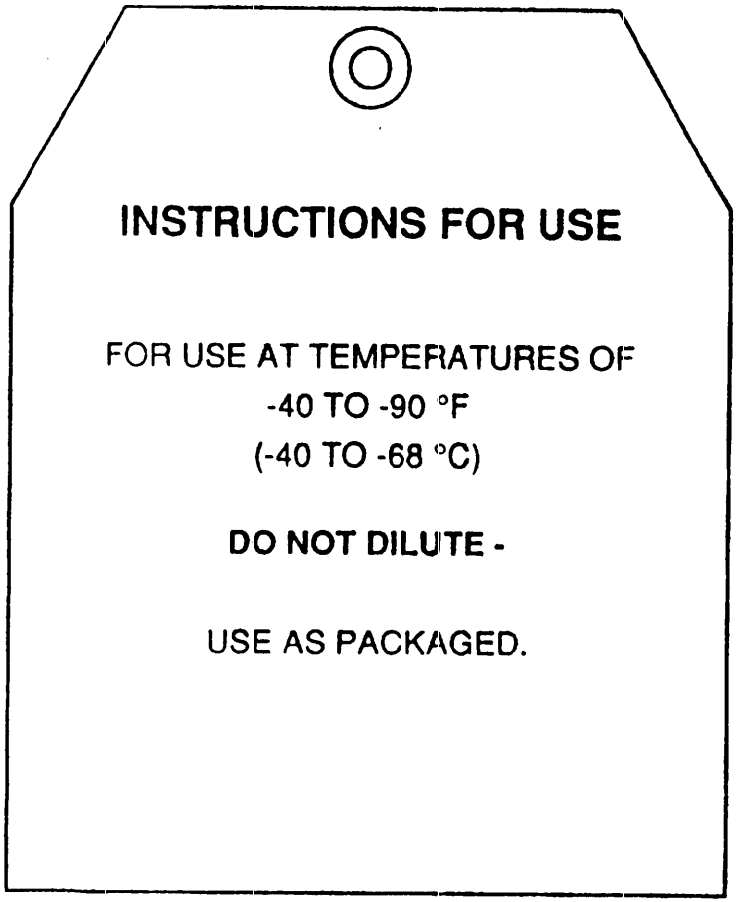
Review activities:
Army - AR, AT, MR, SM
DLA - GS

User activities:
Navy - MC, SH

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SIDE 1



SIDE 2

FIGURE 1. Tag.

X-5025

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ATTACH NEAR
RADIATOR FILLER NECK

THIS COOLING SYSTEM CONTAINS
ETHYLENE GLYCOL BASE ANTIFREEZE
U.S. GOVERNMENT MIL-A-11755

DO NOT OVERFILL

DATE INSTALLED _____

SPEEDOMETER MILEAGE _____

QUARTS PUT IN _____

PROTECTS TO _____ °F

INSTALLED BY _____

INSTRUCTIONS FOR USE

FOR USE AT TEMPERATURES OF
-40 TO -90 °F
(-40 TO -68 °C)

DO NOT DILUTE -

USE AS PACKAGED.

FIGURE 2. Label.

X-5026

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
   The submitter of this form must complete blocks 4, 5, 6, and 7.
   The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-11755E	2. DOCUMENT DATE (YYMMDD) 940124

3. DOCUMENT TITLE
ANTIFREEZE, ARCTIC-TYPE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468 (2) DSN 654-3468
c. ADDRESS (Include Zip Code) ARMY BELVOIR RDE CTR TN SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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